



Intelligent Honeypot

ABSTRACT

Cyberattacks have increased in frequency and complexity due to the internet's and digital systems' rapid growth. Because traditional security systems primarily rely on predefined signatures and rules, they frequently fall short in detecting contemporary attacks. In order to solve this issue, this project suggests an Intelligent Honeypot-Based Threat Detection and Behavioral Analysis System that efficiently detects cyberthreats by utilizing artificial intelligence and deception tactics. In order to draw in attackers and malicious software, honeypots are used in this system to mimic weak systems and services. All of an attacker's interactions with the honeypot, including network activity, command execution, and login attempts, are securely captured. The behavior and patterns of the attacker are then examined by applying machine learning techniques to the collected data. The system concentrates on behavior-based analysis rather than just known attack signatures, which aids in identifying unknown and zero-day attacks. The system recognizes suspicious activity, categorizes various attack types, and finds recurring attack patterns. Real-time alerts, attack details, and threat severity levels are shown on an intuitive dashboard. All things considered, this project's combination of artificial intelligence and honeypots shows a clever and proactive approach to cyber security. It is appropriate for contemporary network protection since it strengthens security monitoring, improves early threat detection, and gives security experts a better understanding of attacker behavior.

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